

Written Test 2 Rewrite

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1(a). Suppose that $A \subset \mathbb{R}$ and $|A| < \infty$. Prove that if A is Lebesgue measurable then for every $\varepsilon > 0$ there exists a set G that is the union of finitely many disjoint bounded open intervals such that

$$|A \setminus G| + |G \setminus A| < \varepsilon.$$

Proof.

□

1(b). State and prove Egorov's Theorem.

Statement: Suppose (X, S, μ) is a measure space with $\mu(X) < \infty$. Suppose $f_1, f_2, \dots$ is a sequence of S -measurable functions from X to $\mathbb{R}$ that converges pointwise on X to a function $f : X \rightarrow \mathbb{R}$. Then for every $\varepsilon > 0$, there exists a set $E \in S$ such that $\mu(X \setminus E) < \varepsilon$ and $f_1, f_2, \dots$ converges uniformly to f on E .

Proof. Let $\varepsilon > 0$, and fix $n \in \mathbb{Z}^+$. For $m \in \mathbb{Z}^+$, let

$$A_{m,n} = \bigcap_{k=m}^{\infty} \left\{ x \in X : |f_k(x) - f(x)| < \frac{1}{n} \right\}.$$

Then the definition of pointwise continuity guarantees that

$$\bigcup_{m=1}^{\infty} A_{m,n} = X.$$

Moreover, $A_{1,n}, A_{2,n}, \dots$ is an increasing sequence of sets, so

$$\lim_{m \rightarrow \infty} \mu(A_{m,n}) = \mu(X).$$

Then there exists an $m_n \in \mathbb{Z}^+$ such that

$$\mu(X) - \mu(A_{m_n,n}) < \frac{\varepsilon}{2^n}.$$

Letting n vary now, let

$$E = \bigcap_{n=1}^{\infty} A_{m_n,n}.$$

Then

$$\begin{aligned}
\mu(X \setminus E) &= \mu\left(X \setminus \bigcap_{n=1}^{\infty} A_{m_n, n}\right) \\
&= \mu\left(\bigcup_{n=1}^{\infty} (X \setminus A_{m_n, n})\right) \\
&\leq \sum_{n=1}^{\infty} \mu(X \setminus A_{m_n, n}) \\
&< \varepsilon.
\end{aligned}$$

Now to prove that $f_1, f_2, \dots$ converges uniformly to f on E , let $\varepsilon' > 0$ and let $n \in \mathbb{Z}^+$ such that $1/n < \varepsilon'$. Then $E \subset A_{m_n, n}$, so

$$|f_k(x) - f(x)| < \frac{1}{n} < \varepsilon'$$

for all $k \geq m_n$ and $x \in E$ due to our choice of m_n . □

1(c). Give an example, with justification, of a measure space $(\mathbb{Z}^+, 2^{\mathbb{Z}^+}, \mu)$ such that

$$\{\mu(E) : E \subset \mathbb{Z}^+\} = [0, 1].$$

You must show that it is a measure space.

Proof. □

2(c). State and prove a theorem that justifies the claim *Integrable functions live mostly on sets of finite measure*.

Proof. □

3(b). State and prove a theorem that justifies one of Littlewood's Three Principles. A brief paraphrase of those principles is

- i. Every Lebesgue measurable set is nearly an open set.
- ii. Every Lebesgue integrable function is nearly continuous. (Briefly outline the three major steps of this proof and then provide details for one of those steps.)
- iii. Every pointwise convergent sequence of Lebesgue integrable functions is nearly uniformly convergent.

If you have already addressed one or more of these principles in previous problems, then make a different choice here.

Proof. □

3(c). Suppose λ is Lebesgue measure on $\mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ is a Borel measurable function such that $\int_{\mathbb{R}} |f| d\lambda < \infty$. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$g(x) := \int_{(x-1, x+1)} f d\lambda.$$

Prove that g is uniformly continuous on $\mathbb{R}$.

Proof. □